

An Open Software Development-based Ecosystem of R Packages for Metabolomics Data Analysis

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In a nutshell

- Powerful and flexible infrastructure for efficient analysis of large-scale MS data
- Open collaborative development of software for MS data analysis:
 - Well documented and thoroughly tested
 - Long-term support and maintenance
- Modular package ecosystem to allow creation of custom analysis workflows
- Tightly integrated into 

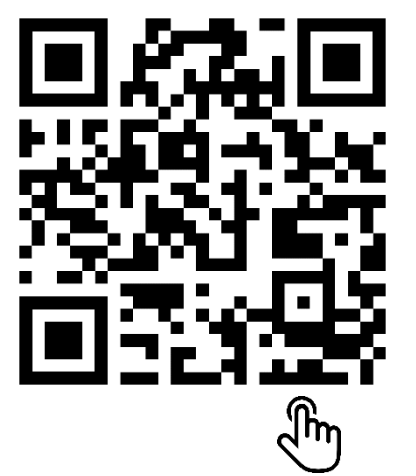
Get in touch/contribute



Poster on [Zenodo](#)

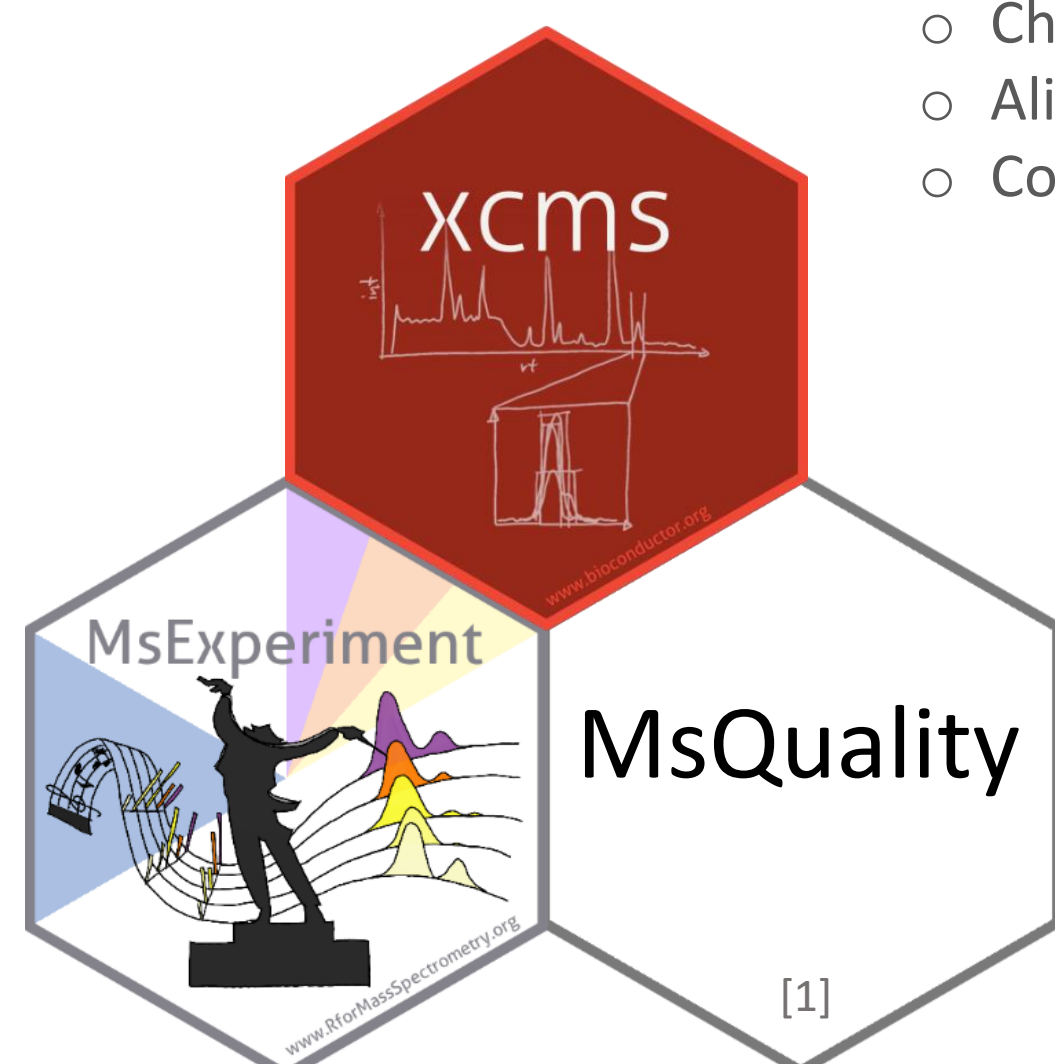


... to see it in action



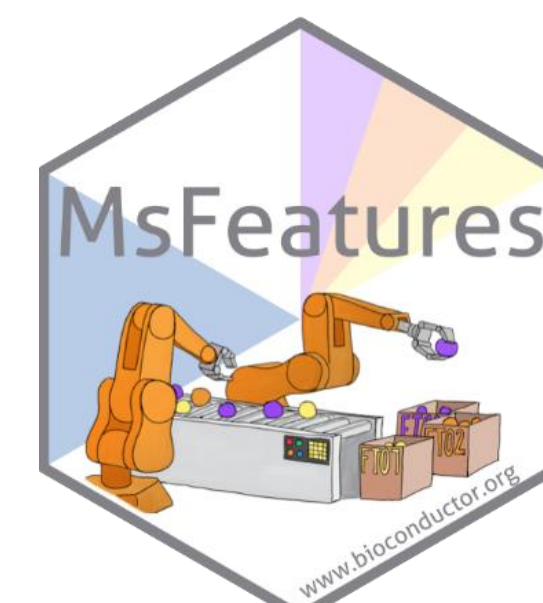
LC-MS(/MS) preprocessing

- Quality assessment
- Chromatographic peak detection
- Alignment
- Correspondence

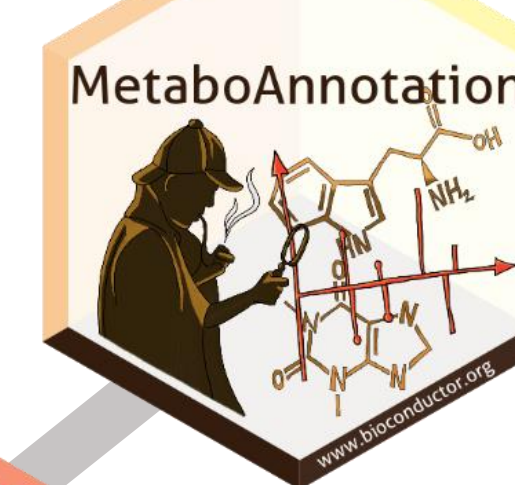


Compounding

- Group adducts
- Group isotopes



Annotation [2]



- MS1 and MS2-based annotation
- Fragment spectra matching

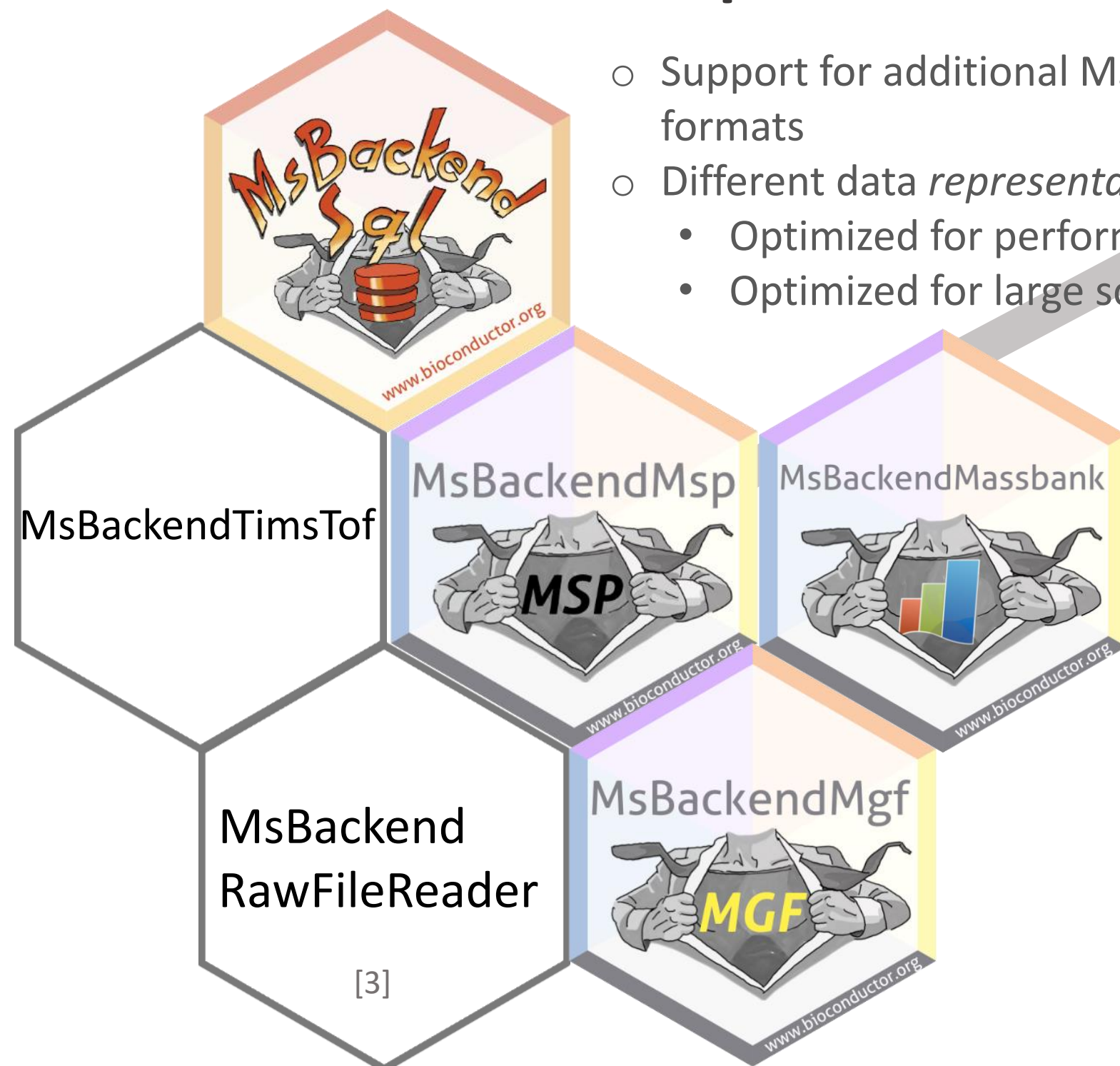
Reference libraries

- In-house databases through CompoundDb
- MassBank through Bioconductor's AnnotationHub
- Import GNPS or others from MGF, MSP files.

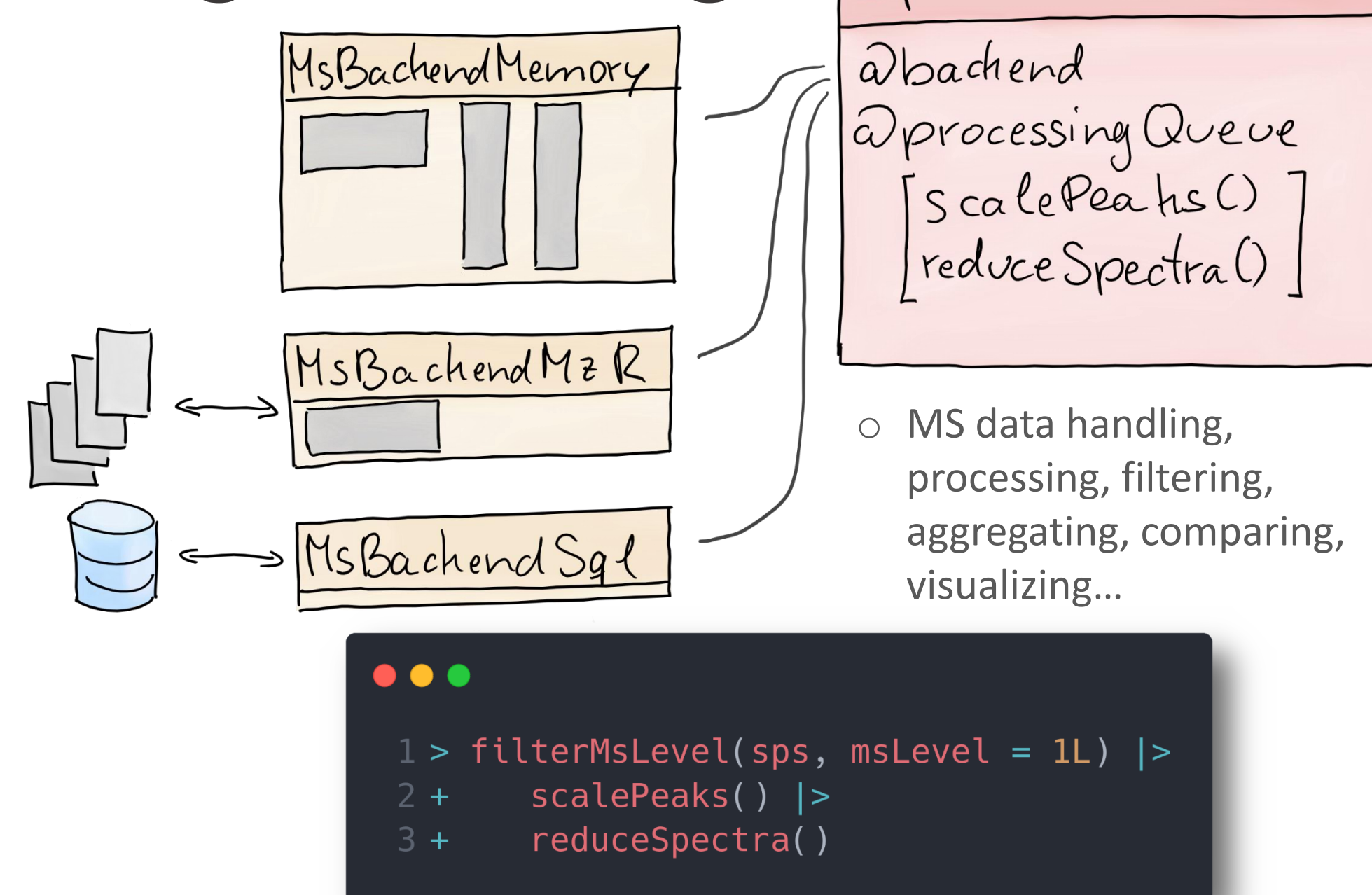


Data I/O and representation

- Support for additional MS data file formats
- Different data representations:
 - Optimized for performance
 - Optimized for large scale data



Design and usage



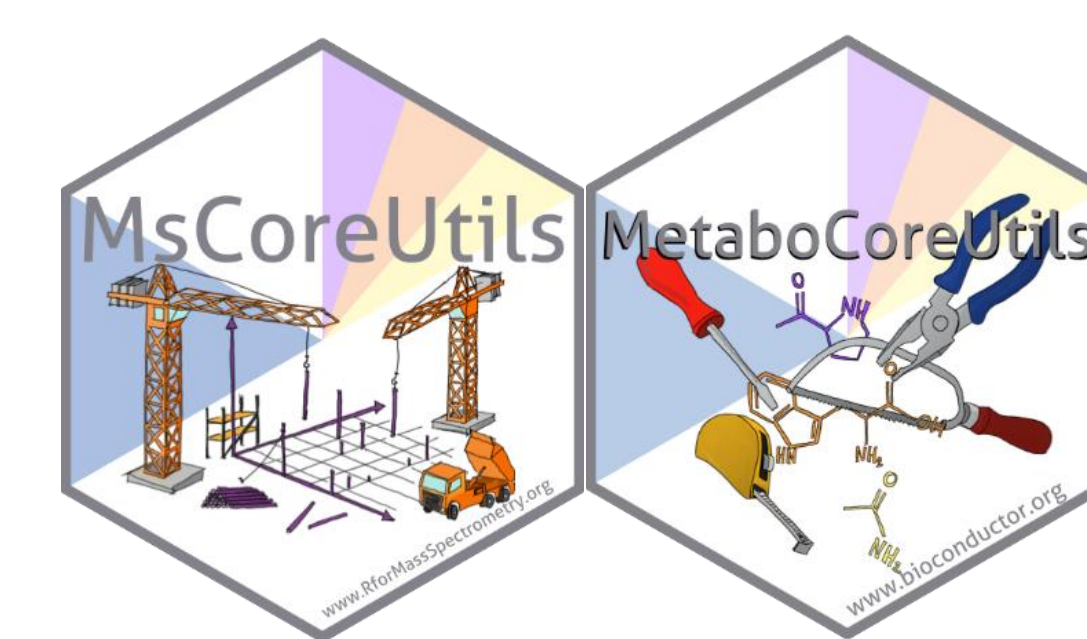
Integration with external tools



- Integrate/use mstchms-based similarity scores for fragment spectra
- Query MS data using the MassQL query language in R.



Core low-level functions



- Spectra similarity scores
- Peak alignment
- Handling chemical formulas
- Isotope peak prediction
- Adduct/exact mass calculations
- etc

Outlook

- Continue integration of R/Bioconductor packages and external software
- Emphasis on seamless integration to improve user experience
- Open for contributions from the community -> reach out via github issue or email

- References**
- [1] MsQuality – an interoperable open-source package for the calculation of standardized quality metrics of mass spectrometry data. Naake T et al. Bioinformatics 2023. <https://doi.org/10.1093/bioinformatics/btad618>
 - [2] A Modular and Expandable Ecosystem for Metabolomics Data Annotation in R. Rainer J et al. *Metabolites* 2022. <https://doi.org/10.3390/metabo12020173>
 - [3] The rawrr R Package: Direct Access to Orbitrap Data and Beyond. Kockmann T et al. Journal of Proteome Research 2021. <https://doi.org/10.1021/acs.jproteome.0c00866>
 - [4] matchms – processing and similarity evaluation of mass spectrometry data. Huber F et al. JOSS 2020. <https://doi.org/10.21105/joss.02411>
 - [5] A Universal Language for Finding Mass Spectrometry Data Patterns. Jarmusch AK et al. bioRxiv 2022. <https://doi.org/10.1101/2022.08.06.503000>